

Python Crash Course

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Outline

- 1 Downloads
- 2 Setup
- 3 Sample Analysis
- 4 GitHub Collaboration

Optionals are not necessary but really nice to have, I will talk a lot about those.

- Download Python 3.x from <https://www.python.org/downloads/>. Click "Add Python to PATH."
- Optional: Download Visual Studio Code from <https://code.visualstudio.com/download>
- Optional: Download Git from <https://git-scm.com/downloads>. Make sure to click "Git Bash" option
- Optional: Create a GitHub account at <https://github.com>.
- Optional: For Latex
 - Download MikTeX from <https://miktex.org/download>
 - Download Strawberry perl from <https://strawberryperl.com/>

Clone from GitHub

- Open your terminal (or Git Bash on Windows)
- Navigate to where you want to store the course folder: `cd ~/Documents`
- Clone the repository:
 - `git clone https://github.com/balazs14/python-course.git`
- Navigate into the cloned folder:
 - `cd python-course`
- Now you have the latest version from GitHub with full git history and can contribute changes!

Setup

- Expand the provided PythonCourse.zip file to a location of your choice.
- Cmd prompt at the PythonCourse folder (right click on folder, select "Open with cmd"). For macOS, open terminal and cd to the PythonCourse folder.
- type: `"python --version <Enter>"` (or try `python3 --version <Enter>` on macOS), you should see the Python version you installed. If you don't, ask AI until you do (reinstall python for example).
- type: `"python setup_env.py <Enter>"`
- At the end of the script, it says "Next steps: Activate the environment" and tells you what to type. Type that.
- type: `"jupyter notebook <Enter>"`. This should open a browser window with the Jupyter interface. This is the basic setup, we can work from here, but I am showing you a better way too.

- Open VSCode
- Open folder: **File** → **Open Folder...** and select the unzipped PythonCourse folder.
- On the left side you find the file explorer. Click `python_intro_stats_notebook.ipynb`
- If VS Code asks to install recommended extensions, click **Yes**.
- If VS Code asks what kernel to use, always choose "python virtual environment" and click the one called "venv".
- If VS Code complains about "could not find python interpreter for jupyter", just click away the popup.

Work in Notebooks

- `python_intro_stats_notebook.ipynb`
- `numpy_pandas_guided_tour_finance.ipynb`

Sample Analysis Results

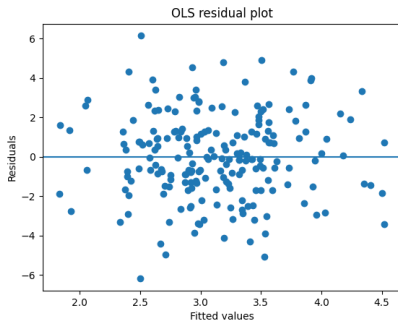


Figure: Some Fancy Analysis

As we can see in Figure 1, our publication deserves a lot of citations.

Collaborating on LaTeX via GitHub

- **Initialize repository:** Create a repo on GitHub, clone it locally
- **Create a branch** for your feature: `git checkout -b feature-name`
- **Edit your .tex file(s)** and commit regularly:
 - `git add <file>`
 - `git commit -m "Clear message about changes"`
 - `git push origin feature-name`
- **Open a Pull Request (PR)** on GitHub for review
- **Resolve conflicts** if others edited the same lines
- **Merge** after approval

Tips for LaTeX Collaboration

- **One sentence per line** in your .tex files—makes diffs clearer and conflicts easier to resolve
- **Commit frequently**—small, logical changes are easier to review and merge
- **.gitignore**—exclude build artifacts:
 - *.aux, *.log, *.pdf, *.fdb_latexmk, *.fls, *.nav, *.snm, *.toc
- **Communicate**—discuss large structural changes in issues before editing
- **Use descriptive branch names**: fix-typos, add-appendix, reformat-intro

Questions?